
Risk-Controlled Prediction Sets for Entity Linking

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Abstract

Entity linking (EL), the task of grounding free-text entity mentions to identifiers in structured resources that formally organize a domain’s knowledge, is a foundational task across many fields and is particularly important for biomedical text mining and knowledge extraction. Existing biomedical EL models typically return a list of ranked candidate entities, leaving investigators to choose an arbitrary cutoff or rely on uncalibrated model scores to determine which candidates to keep. As a result, these models provide no finite-sample guarantee that the correct entity is contained in the retained candidate list. To address this gap, we propose RCPS-EL, a model-agnostic framework for constructing conformal candidate sets for biomedical EL. Building on risk-controlled prediction sets, RCPS-EL converts ranked candidate outputs into calibrated sets that provide user-specified coverage guarantees while controlling the probability of excluding the correct grounding entity. We evaluated RCPS-EL across multiple biomedical EL methods and benchmarking datasets. Our results indicate that RCPS-EL achieves target coverage while producing more compact candidate sets, enabling downstream users to reason about EL uncertainty in a statistically principled way. RCPS-EL is available as an open-source Python package at https://github.com/buzgalbraith/rcps_el.

1 Introduction

Entity linking (EL) is the task of identifying entity mentions in free text and connecting them to unique identifiers in structured resources. In the biomedical domain, this involves grounding mentions of genes, drugs, and other entities to resources such as UMLS [1]. This process creates semantic links allowing automated systems to reason over different sources, underpinning many knowledge

integration and text-mining pipelines. Existing work in biomedical EL falls into two primary categories: (1) Methods that leverage lexical similarity for candidate retrieval (e.g. Gilda [2]), and (2) Methods that leverage learned entity representations during candidate retrieval, including dense-retrieval and generative methods (e.g. SapBERT [3], KRISSBERT [4], GenBioEL [5]). Regardless of the approach, these systems typically return a ranked list of candidates without formal assurance that the correct grounding is contained in the returned set. Separately, conformal prediction [6] has emerged as a principled framework for constructing set-valued predictors with distribution-free coverage guarantees; Risk-Controlling Prediction Sets (RCPS) [7] further extend this to general monotone risk functions. However, to our knowledge, these methods have yet to be applied to EL.

To address this gap, we present Risk-Controlled Prediction Set Entity Linking (RCPS-EL), a model-agnostic framework for constructing conformal candidate sets for biomedical EL that provide user-specified coverage guarantees at a controlled error rate. RCPS-EL is a modular and extensible framework supporting multiple conformal score and loss functions. We then evaluated RCPS-EL across multiple biomedical EL methods and benchmarking datasets. Initial results indicate that RCPS-EL achieves target coverage while producing more compact candidate sets, enabling downstream users to reason about EL uncertainty in a statistically principled way.

2 Methods

Let $D = (X, Y)$ denote a generic data unit from the target population, where $X \in \mathcal{X}$ is a free-text entity mention and $Y \in \mathcal{Y}$ is its corresponding identifier in a structured resource. Given a pre-trained EL model f , let $T_q : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$ denote the set-valued prediction function that maps a mention X to a subset $T_q(X) \subseteq \mathcal{Y}$ of candidate identifiers, with larger values of the tuning parameter q yielding smaller candidate sets. We denote by q_0 the default parameter value used by the pre-trained EL system, so that $T_{q_0}(X)$ coincides with the candidate set returned by the original method. In practice, this default set may be conservative, i.e., it can be too large for efficient downstream curation use. We therefore seek a calibrated value q^* yielding smaller candidate sets than the default while satisfying a user-specified risk constraint over the target population. Formally, let $R(q) = \mathbb{E}\{L(Y, T_q(X))\}$ denote the population risk of the candidate set $T_q(X)$, where $L(\cdot, \cdot)$ is a user-specified loss function, such as the binary miscoverage loss $L(Y, T_q(X)) = 1\{Y \notin T_q(X)\}$. We seek a calibrated tuning parameter that minimizes the expected candidate set size, subject to controlling the relative increase in risk compared with the default EL output:

$$q^* \in \arg \min_{q \in \mathcal{Q}} \mathbb{E}\{|T_q(X)|\} \quad \text{subject to} \quad \frac{R(q) - R(q_0)}{R(q_0)} \leq \alpha.$$

Since $R(q)$ is unknown, RCPS-EL uses a held-out calibration sample to select a data-adaptive $\hat{q}$ such that, with probability at least $1 - \delta$ over the calibration sample, $\{R(\hat{q}) - R(q_0)\}/R(q_0) \leq \alpha$. The resulting risk-controlled EL candidate set is then $T_{\hat{q}}(X) = \{y \in f(X) : s(X, y) \geq \hat{q}\}$, where $s(X, y)$ is the score assigned by the pre-trained EL model to candidate identifier y for mention X .

3 Results and Discussion

We benchmark RCPS-EL on two case studies covering a lexical and a representation-learning biomedical EL method. Figure 1a shows the application of RCPS-EL to the Gilda method for controlling risk on the BioCreative VI Track 1 (Bio-ID) benchmark dataset [8]. Bio-ID is a widely used biomedical EL benchmark and was previously used to benchmark Gilda. Here we use binary miscoverage loss and multiple conformal score functions to narrow Gilda’s prediction set size at various relative risk tolerances. We can see across score functions that RCPS-EL is able to narrow the candidate set size while staying near the desired level of risk tolerance. Figure 1b demonstrates using RCPS-EL to control KRISSBERT model risk on the BioCreative V Chemical Disease Relation (CDR) dataset [9], used for benchmarking by Kartchner et al. [10]. Here we use KRISSBERT to retrieve 20 initial candidates for each mention, then use the KRISSBERT internal score as a conformal score, and explore performance with various hits@ K loss functions. We can again see that the method is able to narrow the prediction set across K -values, while staying below desired risk levels.

These case studies demonstrate the applicability of conformal methods to biomedical EL, and the flexibility of the RCPS-EL framework across both base model types and conformal score/loss

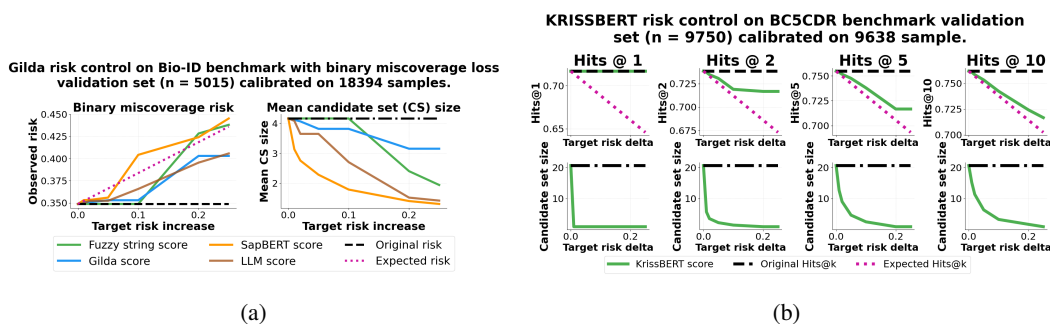


Figure 1: RCPS-EL benchmarking (a) Risk control of Gilda method on Bio-ID benchmark, across score functions and risk tolerances. (b) Risk control of KRISSBERT method on BC5CDR benchmark, with KRISSBERT score across hits@ K losses.

functions. Directions for future work include incorporating loss functions based on ontology tree distance, as well as selective classification and group fairness based approaches.

Acknowledgments

This work was partially supported by grant HR00112220036 under the DARPA ASKEM and ARPA-H BDF programs.

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Appendix

Appendix A: Implementation Details

RCPS-EL is modular with respect to the choice of score function and loss function. We implement several score functions for ranking candidate identifiers, including lexical edit distance, Gilda’s internal score, KRISSBERT’s internal score, cosine similarity between SapBERT embeddings of the mention and candidate identifier, and scores estimated by a large language model. We also implement binary miscoverage and hits@ K losses, which allow the calibration target to be defined in terms of inclusion of the true identifier or retrieval performance among the top-ranked candidates. Throughout calibration and prediction, RCPS-EL restricts attention to mentions with more than one candidate grounding, since a singleton candidate set cannot be reduced without returning the empty set.